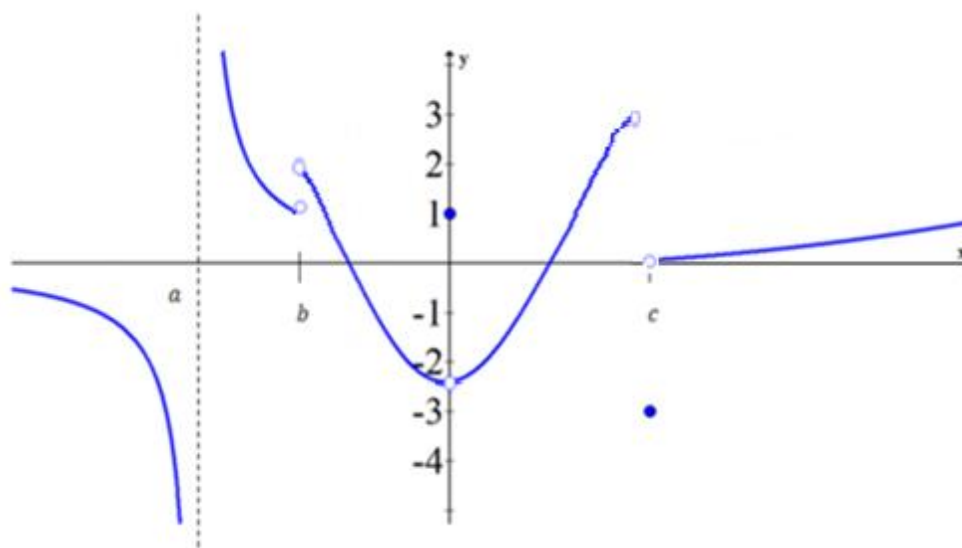


2.2 Limits (An Intuitive Approach) Worksheet

1. Consider the following graph of the function $f(x)$. Determine each of the following limits.



a) $f(0)$

b) $\lim_{x \rightarrow 0^-} f(x)$

c) $\lim_{x \rightarrow 0^+} f(x)$

d) $\lim_{x \rightarrow 0} f(x)$

e) $f(c)$

f) $\lim_{x \rightarrow c^-} f(x)$

g) $\lim_{x \rightarrow c^+} f(x)$

h) $\lim_{x \rightarrow c} f(x)$

i) $f(a)$

j) $\lim_{x \rightarrow a^-} f(x)$

k) $\lim_{x \rightarrow a^+} f(x)$

l) $\lim_{x \rightarrow a} f(x)$

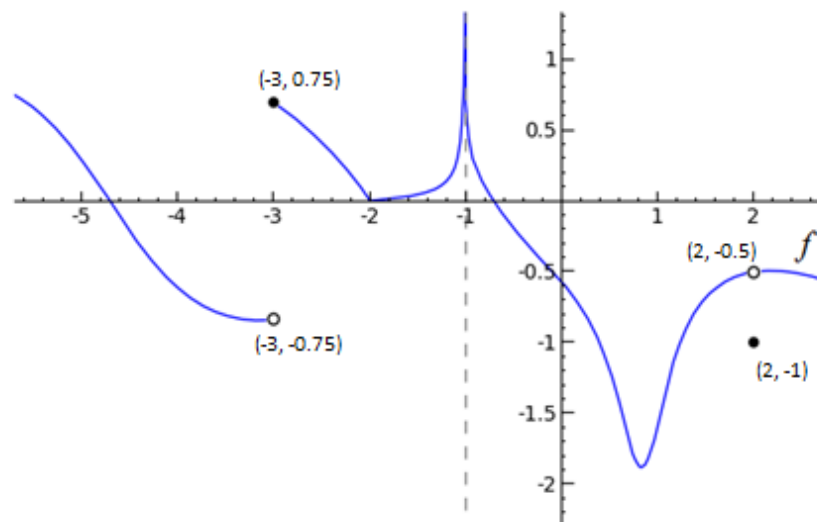
m) $f(b)$

n) $\lim_{x \rightarrow b^-} f(x)$

o) $\lim_{x \rightarrow b^+} f(x)$

p) $\lim_{x \rightarrow b} f(x)$

2. Consider the following graph of the function $g(x)$. Determine each of the following limits.



a) $\lim_{x \rightarrow -3} f(x)$

b) $f(2)$

c) $\lim_{x \rightarrow -1^+} f(x)$

d) $f(-3)$

e) $\lim_{x \rightarrow 2} f(x)$

f) $f(-1)$

g) $\lim_{x \rightarrow -1} f(x)$

h) $\lim_{x \rightarrow -3^-} f(x)$

i) $\lim_{x \rightarrow -2} f(x)$

3. Sketch the graph of an example of a function f that satisfies all of the given conditions:

$$\lim_{x \rightarrow 0^-} f(x) = 1, \quad \lim_{x \rightarrow 0^+} f(x) = -1, \quad \lim_{x \rightarrow 2^-} f(x) = -\infty, \quad \lim_{x \rightarrow 2^+} f(x) = 1, \quad f(2) = 1, \quad f(0) \text{ is undefined}$$